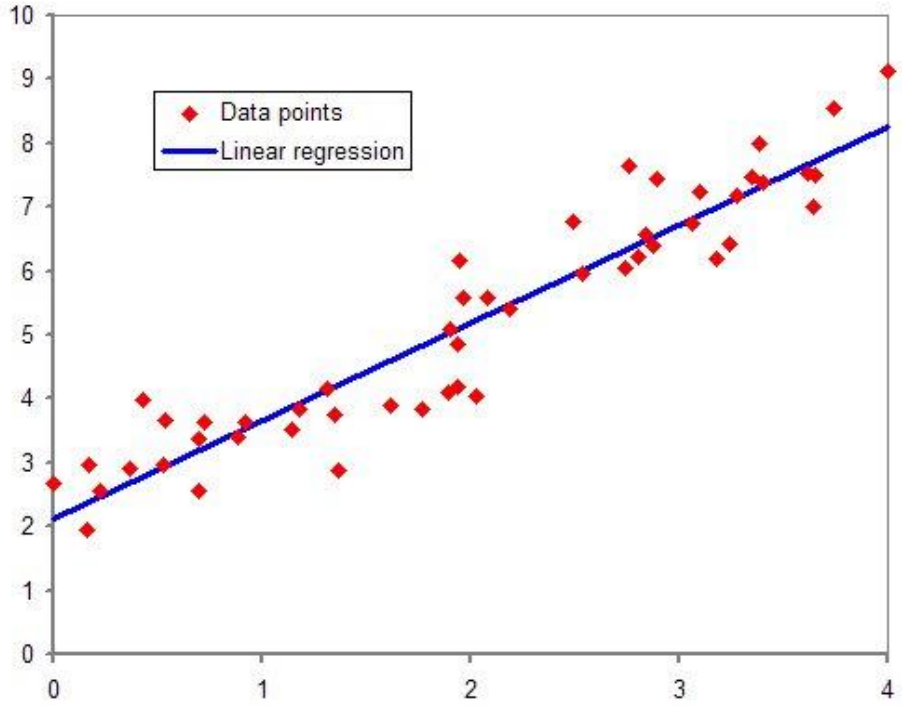
Machine Learning 24CSA301

UNIT II

LINEAR REGRESSION, BIAS-VARIANCE,

Regression

- Linear Regression is a fundamental statistical method used to define linear relationship between a dependent and one or more independent variables.
- The most common form of regression analysis is linear regression, in which one finds the line (or a more complex linear combination) that most closely fits the data according to a specific mathematical criterion.
- This allows the researcher to estimate the value of the dependent variable when the independent variables take on a given set of values.



Linear Regression

Dependent and Independent variable

Expression	Independent	Dependent
$y = 3 + 2x$	x	y
$y = x^2 - 2x$	x	y
$z = 5x^2 + 8y^3$	x, y	z

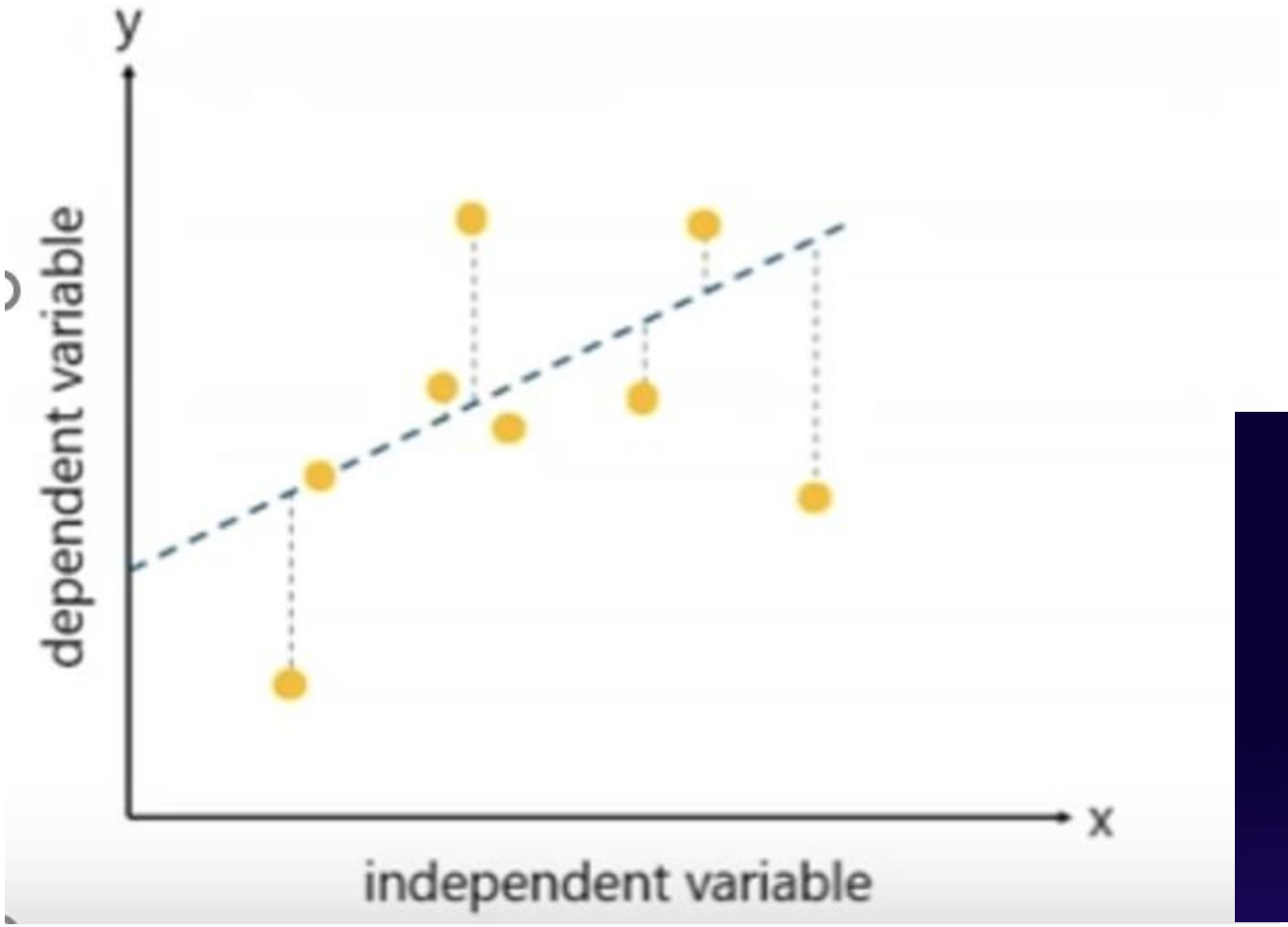
Regression:

Modeling a relationship between ***dependent*** and ***independent*** variables for ***prediction***.

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- *Linear Regression*
  - *Simple Linear Regression* or *Univariate Linear Regression*.
    - Only one independent variable
  - *Multiple Linear Regression* or *Multivariate Linear Regression*.
    - More than one independent variable
- *Non-linear regression*
  - commonly used for more complicated data sets in which the dependent and independent variables show a nonlinear relationship

Linear Regression



$$Y = mX + b$$

- Y represents the dependent variable
- X represents the independent variable
- m is the slope of the line(how much Y changes for a unit change in X)
- b is the intercept (the value of Y when X is 0)

Example

x	y
17	94
13	73
12	59
15	80
16	93
14	85
16	66
16	79
18	77
19	91

$$y = mx+c$$

$$\text{where } m = \sum \frac{(x-\bar{x})(y-\bar{y})}{(x-\bar{x})^2}$$

Applications

- Market Analysis
- Financial Analysis
- Environmental Health

Advantages

- Simple to implement and easier to interpret the coefficients
- If the relationship between the independent variable and dependent variable is linear, the algorithm is best to use because of its less complexity to other algorithms

Drawbacks

- Limited to linear relationships
- Independence presumptions: (The **errors (residuals)** of different observations are independent of each other, observations should not be related to each others)
- Sensitivity to exceptions

Bias and Variance

- Concepts in Machine Learning because they determine **how well a model learns from training data and performs on unseen data**

Total error of a model

- Total Error=Bias^2+Variance+Irreducible Error

Bias

- Bias is the error caused because the learning algorithm makes overly simple assumptions about the data.
- A model with high bias cannot capture the true relationship between the input and output variables.

Characteristics of High Bias/Underfitting

- Oversimplified model
- Cannot learn complex patterns
- High training error
- High testing error
- Underfitting occurs

Causes of High Bias

- Too few features
- Very simple model
- Too much regularization
- Insufficient training

Eg: Linear Regression for nonlinear data

Variance

- Variance measures **how much the model's predictions change when trained on different training datasets**.
- A high variance model memorizes the training data instead of learning general patterns.

Characteristics High Variance Model

- Very complex
- Fits training data perfectly
- Very low training error
- High testing error
- Overfitting

How to Reduce Variance

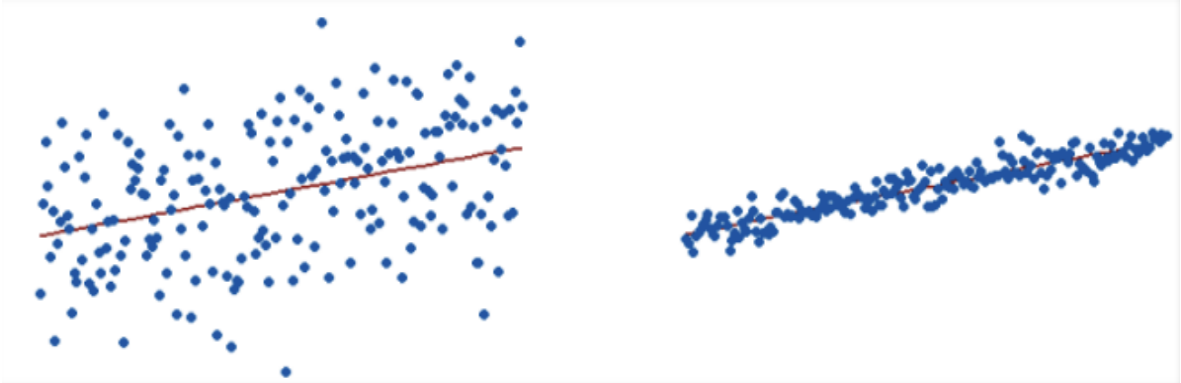
- Increase training data.
- Reduce model complexity.
- Use regularization (L1/L2).
- Perform feature selection.
- Apply cross-validation.
- Use ensemble methods like Random Forest or Bagging.

R-squared: R^2

- A statistical measure that shows how well the independent variable explains the variation of the dependent variable.
- Also called the **Coefficient of Determination**
- It's a score that reflects how well the data fit the regression model, with a value of 1 indicating a perfect fit and 0 indicating no predictive power.

Significance

- The higher the model's R-Squared value, the better the regression line fits the data.
- So if the model value is close to 0, then the model is not a good fit.
- A good model should have an R-squared greater than 0.8.
- Note- As the R squared value increases, the difference between actual and predicted values decreases.



Formula

R^2 is calculated as

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

where:

- SS_{res} = Residual Sum of Squares (unexplained variation)
- SS_{tot} = Total Sum of Squares (total variation in the data)

Residual Sum of Squares (SSE)

$$SS_{res} = \sum (y_i - \hat{y}_i)^2$$

- y_i = Actual value
- $\hat{y}_i$ = Predicted value

It measures the prediction error.

Total Sum of Squares (SST)

$$SS_{tot} = \sum (y_i - \bar{y})^2$$

- $\bar{y}$ = Mean of all actual values

It measures the total variation present in the data.